Statistics & Error Analysis: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Mean and standard error. Return →

The mean is 10.0. The squared deviations are 0.01, 0.01, 0.04, 0, 0.04, summing to 0.10. The sample standard deviation is $\sqrt{0.10/4} = 0.158$. The standard error is $0.158/\sqrt{5} = 0.071$. **Check:** the mean is more precise than a single reading.

Solution 2 — Median versus mean. Return →

The mean is 22.4 and the median is 3. The median is more representative because the value 100 is an extreme outlier that pulls the mean upward. **Check:** four of the five values are near 3.

Solution 3 — Sum of uncertainties. Return →

The sum is 15.0 cm. The uncertainty is $\sqrt{0.2^2 + 0.1^2} = 0.224$ cm, so 15.0 ± 0.2 cm. **Check:** independent absolute uncertainties add in quadrature, not linearly.

Solution 4 — Product uncertainty. Return →

The area is 50.0 cm². The fractional uncertainty is $\sqrt{(0.1/10)^2 + (0.1/5)^2} = 0.0224$, so the absolute uncertainty is $50.0 \times 0.0224 = 1.12$ cm². Report 50.0 ± 1.1 cm². **Check:** the shorter side contributes more because its fractional uncertainty is larger.

Solution 5 — Power law. Return →

$y = 8.00$. The fractional uncertainty is $3(0.02/2.00) = 0.03$, so $\sigma_y = 0.24$. Report 8.00 ± 0.24. **Check:** cubing triples the fractional uncertainty.

Solution 6 — Weighted mean. Return →

The weights are $1/0.2^2 = 25$ and $1/0.5^2 = 4$. The weighted mean is $(25 \times 10.0 + 4 \times 10.5)/29 = 10.069$. The uncertainty is $1/\sqrt{29} = 0.186$. Report 10.07 ± 0.19. **Check:** the more precise measurement receives much more weight.

Solution 7 — More readings. Return →

We need $0.3/\sqrt{n} \leq 0.05$, so $\sqrt{n} \geq 6$ and $n \geq 36$. **Check:** reducing the standard error by a factor of 6 requires 36 times as many readings.

Solution 8 — Systematic offset. Return →

No. Averaging reduces random scatter, not a fixed offset. The ruler's zero error must be calibrated and subtracted from every reading, or included as a systematic uncertainty. **Check:** ten wrong zeros still give a wrong zero.

Solution 9 — Magnitude uncertainty. Return →

$\sigma_m \approx 1.086 \times 0.05 = 0.054$ mag. **Check:** a 1% flux error corresponds to about 0.011 mag, so 5% gives about 0.05 mag.

Solution 10 — Error budget. Return →

The combined uncertainty is $\sqrt{0.03^2 + 0.04^2 + 0.02^2} = 0.054$ mag. Calibration dominates because 0.04 mag is the largest single term. **Check:** improving the random noise alone would not reduce the total much unless calibration is also improved.